MOSE (MOSES) IM

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RESEARCH INTERESTS

Experimental Phonetics; Laboratory Phonology; Speech Timing and Prosody; Computational Modelling; Phonetics–Phonology Interface

EDUCATION

The University of Edinburgh **Sep 2020 - Jun 2026**
MA (Hons), Linguistics and English Language Edinburgh, UK

- First Class Honours (undergraduate degree)
- Dissertation: Modelling the Scottish Vowel Length Rule using the XT/3C articulatory-planning framework
- Supervisors: Prof. Alice Turk and Dr Benjamin Elie

RESEARCH & PROJECT EXPERIENCE

Modelling the Scottish Vowel Length Rule using the XT/3C articulatory-planning framework **Sep 2025 - Apr 2026**
Honours Dissertation Edinburgh, UK

- Investigated whether SVLR-related differences in /u/ and /ai/ vowel duration and quality could emerge from phonetic-planning cost optimisation within a single-category account, rather than requiring distinct short and long phonological categories.
- Ran **Python-based XT/3C and SPARC** optimisation experiments on an **HPC cluster** using Grid Engine job submissions and shell scripting, systematically comparing CTC/PCA and phoneme-level recognition settings.
- Found positive associations between recognition weighting and /ai/ F1 maxima under CTC + PCA=4, while /u/ duration effects remained inconsistent, highlighting limitations of the single-category implementation.

Phonetic Realisation of Tone, Vowel Quality and Voice Quality in Dinka **Sep 2024 - Dec 2024**
(Phonetics and Lab Phonology - Course Project) Edinburgh, UK

- Conducted **acoustic analyses** of 104 target items from four Dinka speakers using Praat and R, measuring F0, F1/F2, spectral tilt and vowel duration to investigate the phonetic realisation of tone and voice-quality contrasts.
- Normalised and visualised acoustic measurements using z-scores, vowel plots and error bars; identified tone-related F0 differences and greater spectral tilt in breathy phonation, while evaluating speaker variability and inconsistent durational patterns.

Small-Vocabulary Speech Recognition with HTK **Sep 2025 - Dec 2025**
(Speech Processing - ASR Course Project) Edinburgh, UK

- Screened a speech corpus to retain 457 speakers, cleaned transcription labels, and trained and evaluated an **HMM-based digit recogniser using HTK** and provided 39-dimensional MFCC features.
- Systematically evaluated the effects of microphone mismatch, accent variation and training-speaker counts on word error rates, identifying diminishing gains from additional training speakers and examining limitations of generalisation in restricted-vocabulary recognition.

Do Scottish English Speakers Perceive SVLR-Related Length Contrasts as Phonemic or Allophonic? **Sep 2025 - Dec 2025**
(Online Experiments for Language Scientists - Course Project) Edinburgh, UK

- Developed an experimental design to investigate whether Scottish English listeners perceive SVLR-related vowel length differences as phonemic or allophonic, adapting an existing identification paradigm with a seven-step BROOD–BREWED duration continuum and sentence-context manipulation.
- Implemented a **two-block jsPsych experiment** with randomised trials, response-time and confidence logging, and headphone checks; evaluated methodological limitations and proposed participant-selection and data-quality criteria. No participant data were collected.

Bayesian Modeling of Language Evolution **Jan 2025 - Apr 2025**
(Simulating Language - Course Project) Edinburgh, UK

- Used provided Python models to investigate how initial-language structure and population turnover influence the emergence of compositionality through repeated simulations of language learning and cultural transmission.
- Examined transmission bottlenecks in separate **Bayesian iterated-learning simulations**, visualising convergence across generations and interpreting the effects of population turnover and bottleneck size in relation to model assumptions and limitations.

Coarticulatory Pressures Interfere with the Learning of Vowel Harmony **Jan 2025 - Apr 2025**
Mock research article (Universals of Language - Course Project) Edinburgh, UK

- Collaborated in a three-person team (one MSc and two Honours students) to develop a research question examining whether vowel-harmony learning preferences reflect a substantive learning bias or coarticulatory pressures mediated by subvocal rehearsal.
- Co-designed an artificial-language learning study with distraction and novel-suffix production conditions, specifying hypotheses, experimental factors, stimuli and predicted outcomes; co-authored an IMRD-format mock research article using hypothetical data. No participant data were collected.

Acoustic and Corpus-Based Analysis using the HCRC MapTask Corpus **Sep 2023 - Dec 2023**
(LEL2B: Phonetic Analysis and Empirical Methods - Course Project) Edinburgh, UK

- Applied Praat to HCRC MapTask word-list and dialogue recordings to measure vowel formants, stop VOT and prosodic timing, comparing phonetic patterns across isolated and connected speech.
- Formulated and tested hypotheses about referring-expression form, mention length and disfluency using HCRC MapTask corpus data, spreadsheet-based annotation and statistical tests; identified shorter subsequent mentions and greater disfluency among Guides than Followers.

TEACHING EXPERIENCE

English Instructor **May 2026 - Sep 2026**
Chungdahm Institute (CDI) Incheon, South Korea

- Taught English reading, listening, speaking and writing to school-age learners; planned group lessons, provided feedback and supported school-exam and CSAT preparation.

English Instructor **May 2025 - Aug 2025**
GnB Language Institute Seoul, South Korea

- Delivered English classes across age and proficiency levels, adapting activities and feedback to mixed-ability groups.

Temporary English Teacher (Individual Support) **Jul 2023 - Sep 2023**
Sangam Middle School Seoul, South Korea

- Provided one-to-one English and learning support for a lower-secondary student experiencing significant behavioural and attendance difficulties, helping them re-engage with schoolwork and meet required lesson hours.
- Adapted lesson content and routines in collaboration with school support staff, focusing on relationship-building, emotional safety and gradual re-engagement with learning.

PROFESSIONAL EXPERIENCE

Military Service **Oct 2021 - Apr 2023**
Republic of Korea Army Gyeonggi-do, South Korea

- Served in an assault infantry battalion, developing discipline, resilience and teamwork through training and operational duties.

Administrative Assistant **Jan 2021 - Oct 2021**
Missions Office, Yewon Church Seoul, South Korea

- Handled day-to-day administration and communication in the missions office, including documents, scheduling, event support, liaising with overseas mission fields, and basic translation/interpretation and relief logistics.

ADDITIONAL TRAINING

Transcribing Prosodic Structure of Spoken Utterances with ToBI **Jan 2026**
Massachusetts Institute of Technology (MIT), Independent Activities Period (IAP) Online

Instructors: Stefanie Shattuck-Hufnagel, Alejna Brugos, and Nanette Veilleux

- Participated in online training in MAE-ToBI prosodic transcription, covering intonation, phrasing, prominence, tones and break indices.

AWARDS

Army Soldier Award (Division Commander's Commendation) - ROK Army **April 2023**
Division Commander's Commendation (Special prize, division-wide video competition) - ROK Army **Nov 2022**

SKILLS

Languages & Programming: Python, R, JavaScript, HTML/CSS, Praat scripting
Frameworks & Libraries: NumPy, pandas, matplotlib, tidyverse, ggplot2, jsPsych, PlanArt (XT/3C), Parselmouth
Tools: Praat, Montreal Forced Aligner (MFA), Festival, HTK, HPC (Grid Engine, Bash), Git, GitHub, LaTeX, Unix shell (bash)